



**The University of Azad Jammu and Kashmir,
Muzaffarabad**

Name: Muneeba Arshad

Roll No: 2024-SE-06

Course: Computer Networks

Course Code: SE-2203

Lab NO : 02

Submitted To: Engr. Dr. Asma Javaid

Department Of Software Engineering

✓ **Explore Packet Tracer Interface**

1. Purpose of the Experiment

The purpose of this experiment was to get an initial understanding of Cisco Packet Tracer and its use in computer networking. The following tasks were performed during the lab:

- Become familiar with the Packet Tracer working environment.
- Examine the different categories of devices available in the software.
- Learn how network devices are placed and connected.
- Understand the difference between logical network design and physical device arrangement.
- Explore the basic configuration options available for network devices.
- Observe network communication using Packet Tracer's operating modes.
- Build a small network and examine its connectivity.

2. Background

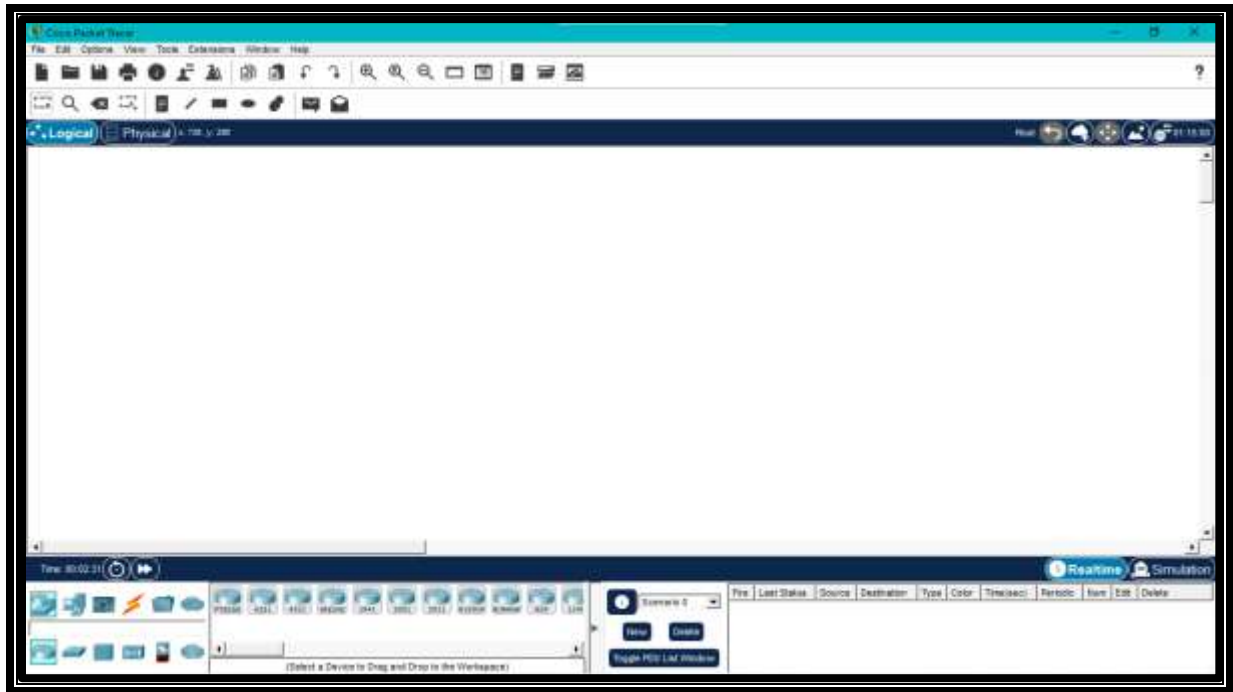
Cisco Packet Tracer is a software-based networking simulator that allows users to build and experiment with networks on a computer. Instead of connecting real routers, switches, and computers, virtual versions of these devices can be used.

This makes Packet Tracer particularly useful for students because networking concepts can be practiced without needing a physical laboratory containing expensive equipment. A user can create a network, assign addresses, connect devices, test communication, and make changes to the design.

The program can be used to practice many networking topics, including IP addressing, switching, routing, VLAN configuration, DHCP, DNS, and troubleshooting.

3. Exploring the Packet Tracer Environment

After opening Packet Tracer, several sections of the interface can be seen. Each section has a specific purpose in creating and managing a network.



3.1 Menu Options

The menu section at the top of the application provides access to different commands and settings.

For example:

- **File** is used for creating, opening, saving, and closing projects.
- **Edit** provides options for modifying the current work.
- **Options** contains program preferences and settings.
- **View** controls how different parts of the interface are displayed.
- **Tools** provides additional utilities for working with the network.
- **Extensions** contains extra features provided by Packet Tracer.
- **Help** provides assistance and information about the application.

3.2 Toolbar

The toolbar contains shortcuts for actions that are commonly required while working on a project.

Some of the frequently used functions are:

- Creating a new project
- Opening an existing project
- Saving a project
- Printing the network design
- Undoing an operation
- Redoing an operation
- Increasing or decreasing the workspace size

These shortcuts make it easier to perform routine tasks without repeatedly opening the menus.

4.1 Logical Workspace

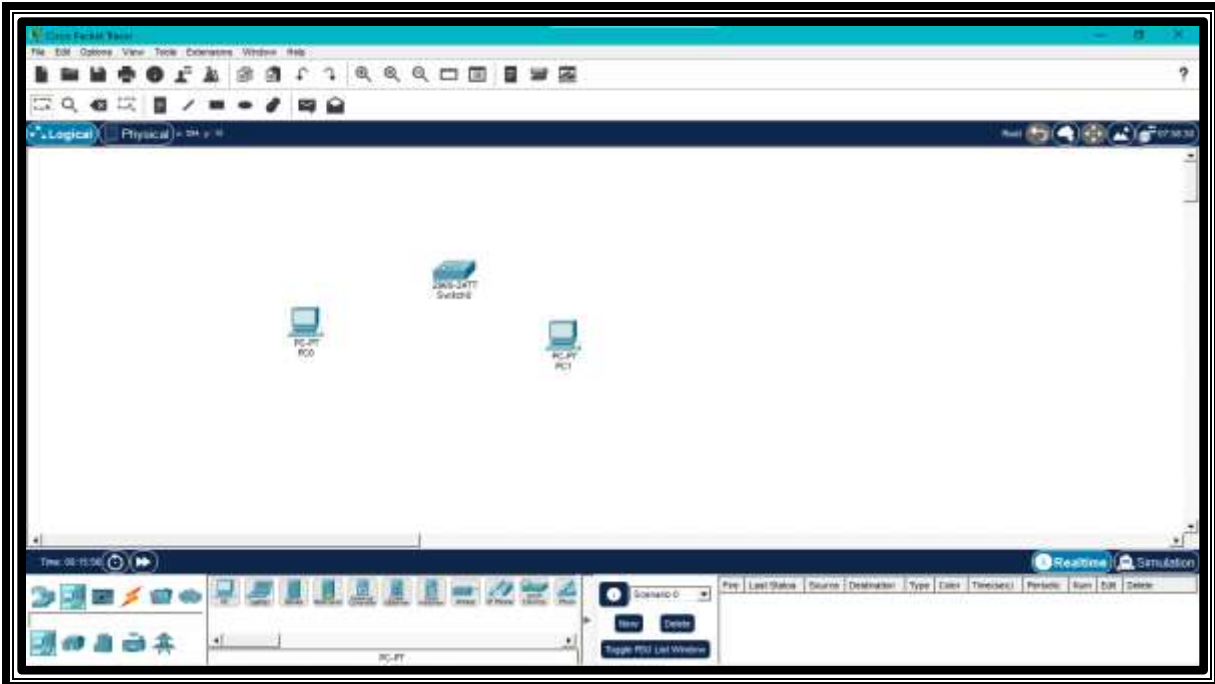
The Logical Workspace is where the actual network topology is normally designed. Devices can be placed on the screen and connected according to the required network structure.

For example, a simple arrangement may look like:

Computer → Switch → Router

In this area, users can add devices, establish links, configure network settings, and test whether devices can communicate.

The main focus of this workspace is the **relationship and connectivity between network devices**.



4.2 Physical Workspace

The Physical Workspace presents the network from a more realistic physical perspective.

Instead of concentrating only on connections, it gives an idea of where equipment might physically be located. Devices can be viewed in areas such as rooms or racks, and their physical connections can be examined.

This view is useful for understanding the relationship between a network's logical design and its actual physical implementation.

5. Devices Available in Packet Tracer

A variety of devices can be selected from the device panel.

5.1 Networking Equipment

Networking equipment is responsible for connecting devices and controlling the movement of data.

Some examples are:

- **Router:** Used to connect separate networks and route packets between them.

- **Switch:** Used to connect devices within a local network.
- **Hub:** Provides a basic method of connecting multiple devices.
- **Wireless Devices:** Provide wireless networking functionality.

5.2 End Devices

End devices are generally the devices that users interact with or that provide network services.

Examples include:

- Computers
- Laptops
- Servers
- Printers
- IP Phones
- Tablets

These devices can be connected to switches, routers, wireless devices, or other network components.

6. Connecting Devices

After placing devices in the workspace, connections can be established between them.

Packet Tracer provides several types of connections, including:

- Copper straight-through
- Copper crossover
- Fiber
- Console
- Serial

The selection of a cable depends on the interfaces and devices involved in the connection.

For this experiment, a **copper straight-through connection** was used to connect the PCs to the switch.

7. Examining a Device

A device can be opened by selecting it from the workspace. A separate window then provides access to different features.

For a PC, the major sections include:

Physical Section

This section represents the physical hardware of the computer. It allows the user to view the components associated with the device.

Config Section

The Config section provides options for basic device and interface configuration.

Desktop Section

The Desktop section contains several utilities that can be used for network-related tasks.

Important utilities include:

- IP Configuration
- Command Prompt
- Web Browser
- Terminal

The Command Prompt is particularly useful for testing connectivity. For example, the following command can be entered:

```
ping 192.168.1.1
```

If the destination device is reachable and correctly configured, the ping operation can return responses.

8. Network Operating Modes

Packet Tracer provides two different ways to observe network activity.

Realtime Mode

Realtime Mode allows the simulated network to operate continuously. It is suitable for normal configuration and testing activities.

When a device sends data, the communication takes place without requiring the user to manually advance through each stage.

Simulation Mode

Simulation Mode provides a closer look at what happens to packets as they move through a network.

Using this mode, individual network events can be examined. It can help users understand packet forwarding, protocol activity, successful delivery, and communication problems.

This feature is especially useful when learning how networking protocols and devices process data.

9. Practical Network Setup

As part of the experiment, a small network was constructed using:

- Two PCs
- One switch
- Two copper straight-through cables

The arrangement was:

PC0 — Switch0 — PC1

The first PC was connected to one port of the switch, while the second PC was connected to another available port.

After making the connections, the status indicators were checked. The link indicators provide a quick visual indication of the state of a connection. Once the connections became operational, the network was ready for further testing.

10. Learning and Observations

The experiment provided practical exposure to several important Packet Tracer features.

It was observed that Packet Tracer makes it possible to construct a network by simply selecting devices and placing them in the workspace. Connections can then be created between the devices using the required cable type.

The experiment also showed that the Logical and Physical workspaces provide different perspectives of the same network. The Logical view concentrates on network topology, while the Physical view gives more attention to the actual placement and hardware arrangement.

The device windows were also examined to understand how configuration and troubleshooting tools can be accessed.

Finally, the difference between normal network operation and packet-by-packet observation was studied using Realtime and Simulation modes.

11. Outcome

The experiment was completed successfully. The basic features of Cisco Packet Tracer were explored, and different types of network devices and connections were identified.

A small network containing two PCs and a switch was also created. The devices were connected successfully, and the status of the network links was observed.

The experiment provided practical familiarity with the basic process of designing and examining a computer network in a simulated environment.

12. Conclusion

This experiment introduced the fundamental features of Cisco Packet Tracer and demonstrated how it can be used as a virtual networking laboratory. The software makes it possible to experiment with network devices and communication without using physical equipment.

During the lab, the interface, device categories, connection methods, configuration windows, and workspace options were explored. A basic PC-to-switch network was also constructed and examined.

